

LECTURELENS

Fully Offline Lecture Copilot Accelerated for Snapdragon PCs

Qualcomm Snapdragon AI Lab Challenge Submission
100% Private • NPU Accelerated • Zero Cloud Dependency



The Problem: Classroom Wi-Fi Deadzones & Cloud Latency



Campus Wi-Fi Deadzones

Overcrowded lecture halls suffer from severe bandwidth throttling or zero cellular coverage, rendering cloud AI tools unusable.



Fast-Paced STEM Lectures

Engineering & STEM students struggle to listen attentively while scribbling complex equations and technical terminology.



Privacy & IP Concerns

Uploading proprietary course lectures or confidential research audio to public cloud APIs violates data privacy policies.

The Solution: LectureLens On-Device AI Copilot

100% Data Privacy

All audio processing, neural transcription, and LLM summary generation happen locally on the laptop with zero cloud requests.

Hexagon NPU Acceleration

Optimized with Qualcomm QNN Execution Provider (HTP) & Foundry Local NPU model endpoints for maximum battery efficiency.

Automated Study Artifacts

Instant conversion of raw classroom speech into executive notes, key term glossaries, interactive quizzes, and flashcards.

Student Workflow: From Audio Stream to Knowledge Mastery

1. Record / Upload

Live mic capture or
WAV/MP3/FLAC file
import

2. Transcribe

Neural Whisper STT with
sub-second chunking

3. Summarize

Executive bullets, key
terms & revision
paragraph

4. Quiz & Flashcards

Interactive multiple choice
Q&A & active recall cards

5. Hybrid Search

Cross-lecture vector
embeddings & BM25
keyword index

System Architecture & On-Device Data Pipeline

Audio Resampler & VAD

16 kHz mono conversion with energy-based voice activity detection.

Whisper STT (QNN/HTP)

ONNX Runtime QNN provider for NPU speech recognition + PyTorch CPU fallback.

Foundry Local LLM

Standardized local NPU chat endpoint + Ollama local CPU fallback.

Hybrid Search Engine

ONNX MiniLM passage embeddings + BM25 keyword index.

Snapdragon NPU Acceleration & Dual Engine Architecture

Qualcomm Hexagon NPU Offloading

Offloads dense matrix multiplications to Hexagon NPU to extend battery life and keep CPU thermals cool during long 2-hour lectures.

ONNX Runtime + QNN Provider

``stt_qnn.py`` connects directly to Qualcomm HTP driver for NPU-accelerated Whisper speech recognition.

Foundry Local NPU LLM Endpoint

``llm_foundry.py`` wraps local ``/v1/chat/completions`` API targeting Snapdragon NPU LLM models.

Graceful CPU Fallback Architecture

Zero-downtime fallback to PyTorch & Ollama CPU engines when QNN hardware is absent on non-Snapdragon dev machines.

Honest & Verified Performance Benchmarks

10% WER

STT Accuracy

Verified against 40-word human TTS reference transcript

4.93 tok/sec

LLM Speed (CPU)

Measured on Ollama phi3:mini local CPU engine

~0.166 RTF

STT Processing

Real-time factor measured for Whisper CPU inference

24 / 24

Unit & Integration Tests

100% test suite pass rate verified with pytest

Strict Offline Guarantee & Zero Cloud Interception

0 Cloud Network Calls

Confirmed by
`scripts/offline\_check.py` socket
guard monkeypatching.

100% Data Residency

Audio, transcripts, notes, and vector
databases remain encrypted on local
disk.

Airplane Mode Ready

Operates reliably without active Wi-
Fi, cellular, or Bluetooth networks.

User Experience: iQOO Cyber Yellow & Obsidian Black UI

High-Contrast Streamlit Web Dashboard

- Inspired by flagship iQOO Monster Cyber Yellow (#FFD100) & Pitch Carbon (#000000) styling.
- 8 Responsive Navigation Tabs: Record, Notes, Quiz, Flashcards, Search, Q&A, Performance, History.
- Real-Time Acceleration Sidebar: Shows active hardware status (QNN NPU vs CPU Fallback).
- Accessible & High-Contrast: Optimized typography for fast classroom readability.



Target Persona & Real-World Educational Impact

STEM Students

Rapid extraction of complex engineering concepts, formulas, and technical glossaries.

Law & Medical Students

High-density active recall quiz generation and automatic flashcard creation.

Research Faculty

Secure, confidential lecture recording and seminar analysis with zero privacy leaks.

Future Roadmap: Extending On-Device Intelligence

1. Speaker Diarization

On-device voice embeddings to separate professor vs student questions in live audio.

2. Multilingual Support

Real-time speech translation for Tamil, Hindi, Spanish, and German lectures.

3. Wi-Fi Direct Sync

Local peer-to-peer note sharing between Snapdragon laptops and tablets without internet.

LECTURELENS

Bringing Private, Offline AI to Every Engineering Classroom.

GitHub Repo: <https://github.com/suvakovan/SNAP-DRAGON->
Qualcomm Snapdragon AI Lab Challenge 2026

THANK YOU! QUESTIONS & LIVE DEMO